

5.2.49

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Question

Construct a pair of tangents to a circle of radius 4 cm from a point P at a distance of 7 cm from its centre O .

Solution

Given:

Let the centre be origin and $\mathbf{P}$ lies on x axis.

$$\mathbf{x}^T \mathbf{x} = 16 \quad (1)$$

$$\mathbf{P} = \begin{pmatrix} 7 \\ 0 \end{pmatrix}, \quad (2)$$

Equation of a line through $\mathbf{P}$ with direction $\mathbf{d}$:

$$\mathbf{x} = \mathbf{P} + t\mathbf{d}, \quad t \in \mathbb{R}. \quad (3)$$

Substitute (3) in (1) : $(\mathbf{P} + t\mathbf{d})^T (\mathbf{P} + t\mathbf{d}) = 16$

$$\Rightarrow (\mathbf{d}^T \mathbf{d})t^2 + 2(\mathbf{P}^T \mathbf{d})t + (\mathbf{P}^T \mathbf{P} - 16) = 0 \quad (4)$$

For tangency, discriminant = 0:

$$\begin{aligned} (2\mathbf{P}^T \mathbf{d})^2 - 4(\mathbf{d}^T \mathbf{d})(\mathbf{P}^T \mathbf{P} - 16) &= 0 \\ \Rightarrow (\mathbf{P}^T \mathbf{d})^2 &= (\mathbf{d}^T \mathbf{d})(\mathbf{P}^T \mathbf{P} - 16) \end{aligned} \quad (5)$$

$$\mathbf{P}^T \mathbf{P} = 49, \quad (6)$$

$$(\mathbf{P}^T \mathbf{d})^2 = 33(\mathbf{d}^T \mathbf{d}). \quad (7)$$

For a unit vector $\mathbf{d}$,

$$\mathbf{d}^T \mathbf{d} = 1 \Rightarrow (\mathbf{P}^T \mathbf{d})^2 = 33 \quad (8)$$

Let

$$\mathbf{M} = \mathbf{P}\mathbf{P}^T \Rightarrow \mathbf{d}^T \mathbf{M} \mathbf{d} = (\mathbf{P}^T \mathbf{d})^2 \quad (9)$$

$$\mathbf{d}^T \mathbf{M} \mathbf{d} = 33, \quad (10)$$

$$\mathbf{M} = \mathbf{P}\mathbf{P}^T = \begin{pmatrix} 7 \\ 0 \end{pmatrix} \begin{pmatrix} 7 & 0 \end{pmatrix} = \begin{pmatrix} 49 & 0 \\ 0 & 0 \end{pmatrix} \quad (11)$$

Solution

Eigen values and Eigen vectors of M:

$$\lambda_1 = 49, \lambda_2 = 0.$$

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (12)$$

$\mathbf{d}$ in orthonormal Eigenbasis : $\mathbf{d}^T \mathbf{d} = 1$

$$\mathbf{d} = \alpha \mathbf{e}_1 + \beta \mathbf{e}_2, \quad \alpha^2 + \beta^2 = 1 \quad (13)$$

$$\mathbf{d}^T \mathbf{M} \mathbf{d} = \lambda_1 \alpha^2 + \lambda_2 \beta^2 \quad (14)$$

$$\Rightarrow \mathbf{d}^T \mathbf{M} \mathbf{d} = 49\alpha^2 + 0 \cdot \beta^2 = 49\alpha^2 \quad (15)$$

$$\text{From (10), } 49\alpha^2 = 33 \Rightarrow \alpha = \pm \frac{\sqrt{33}}{7}. \quad (16)$$

$$\Rightarrow \beta = \pm \frac{4}{7} \quad (17)$$

Solution

Hence $\mathbf{d}$ is:

$$\therefore \mathbf{d} = \begin{pmatrix} \frac{\sqrt{33}}{7} \\ \pm \frac{4}{7} \end{pmatrix} \quad (18)$$

From (4) and (8)

$$\begin{aligned} t &= -\frac{\mathbf{P}^T \mathbf{d}}{\mathbf{d}^T \mathbf{d}} \\ \Rightarrow t &= -\sqrt{33} \end{aligned} \quad (19)$$

Required Points of Contact

$$\mathbf{T} = \mathbf{P} + t\mathbf{d} = \mathbf{P} - \sqrt{33}\mathbf{d} \quad (20)$$

$$\mathbf{P} = \begin{pmatrix} 7 \\ 0 \end{pmatrix}, \mathbf{d} = \begin{pmatrix} \frac{\sqrt{33}}{7} \\ \pm \frac{4}{7} \end{pmatrix} : \quad (21)$$

$$\mathbf{T} = \begin{pmatrix} 7 \\ 0 \end{pmatrix} - \sqrt{33} \begin{pmatrix} \frac{\sqrt{33}}{7} \\ \pm \frac{4}{7} \end{pmatrix} = \begin{pmatrix} \frac{16}{7} \\ \mp \frac{4\sqrt{33}}{7} \end{pmatrix} \quad (22)$$

$$\mathbf{T}_1 = \begin{pmatrix} \frac{16}{7} \\ \frac{4\sqrt{33}}{7} \end{pmatrix}, \quad \mathbf{T}_2 = \begin{pmatrix} \frac{16}{7} \\ -\frac{4\sqrt{33}}{7} \end{pmatrix} \quad (23)$$

Equation of tangents :

$$\mathbf{x} = \begin{pmatrix} 7 & 0 \end{pmatrix}^T + k \begin{pmatrix} \frac{\sqrt{33}}{7} & \frac{4}{7} \end{pmatrix}^T \quad (24)$$

$$\mathbf{x} = \begin{pmatrix} 7 & 0 \end{pmatrix}^T + k \begin{pmatrix} \frac{\sqrt{33}}{7} & -\frac{4}{7} \end{pmatrix}^T \quad (25)$$

C Code

```
#include <stdio.h>
#include <math.h>

// Tangent function to be called from Python
void tangent_points(double r, double px, double py, double *Ax,
    double *Ay, double *Bx, double *By) {
    double ox = 0.0, oy = 0.0;
    double d = sqrt((px - ox)*(px - ox) + (py - oy)*(py - oy));

    if (d <= r) {
        *Ax = *Ay = *Bx = *By = NAN;
        return;
    }

    double r2 = r*r;
    double d2 = d*d;
    double k = r2 / d2;
    double h = (r / d2) * sqrt(d2 - r2);
```

```
1  *Ax = k*px - h*py;  
2  *Ay = k*py + h*px;  
3  *Bx = k*px + h*py;  
4  *By = k*py - h*px;  
5  }
```

```
import numpy as np
import matplotlib.pyplot as plt
from numpy import linalg as LA

# Local imports
from libs.line.funcs import line_gen
from libs.conics.funcs import circ_gen

# --- Circle and external point ---
O = np.array([[0.0], [0.0]]) # Centre
r = 4.0 # Radius
P = np.array([[7.0], [0.0]]) # External point

# --- Tangent point calculation ---
u = P - O
d = float(LA.norm(u))
u_hat = u / d
```

Python Direct

```
# Rotation matrix for 90 degrees
R90 = np.array([[0, -1], [1, 0]])

# Tangent points formula
r2 = r**2
sqrt_term = np.sqrt(d**2 - r**2)

A = 0 + (r2/d)*u_hat + (r/d)*sqrt_term*(R90 @ u_hat)
B = 0 + (r2/d)*u_hat - (r/d)*sqrt_term*(R90 @ u_hat)

# --- Generate required curves ---
x_circ = circ_gen(0, r)
x_tan1 = line_gen(P, A)
x_tan2 = line_gen(P, B)
x_OP = line_gen(0, P)

# --- Plotting ---
plt.plot(x_circ[0, :], x_circ[1, :])
```

```
plt.plot(x_tan1[0, :], x_tan1[1, :], color='orange' )
plt.plot(x_tan2[0, :], x_tan2[1, :], color='green' )
plt.plot(x_OP[0, :], x_OP[1, :], 'k--')

# --- Mark and label points (rounded neatly) ---
points = {'O': O, 'P': P, 'A': A, 'B': B}
for name, pt in points.items():
    x, y = float(pt[0]), float(pt[1])
    plt.scatter(x, y, s=45, zorder=5)
    plt.text(x + 0.25, y + 0.25,
             f'{name}({round(x,2)}, {round(y,2)})',
             fontsize=9, ha='center')
```

```
# --- Axes setup ---
ax = plt.gca()
ax.spines['top'].set_color('none')
ax.spines['right'].set_color('none')
ax.spines['left'].set_position('zero')
ax.spines['bottom'].set_position('zero')

# --- Style ---
plt.axis('equal')
plt.grid(True)
plt.xlabel('$x$')
plt.ylabel('$y$')
plt.title('Tangents from an External Point to a Circle')
plt.legend(loc='upper right')
plt.show()
```

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt
from libs.line.funcs import line_gen
from libs.conics.funcs import circ_gen

# Load the shared library
lib = ctypes.CDLL('./libtangent.so')

# Define argument and return types
lib.tangent_points.argtypes = [
    ctypes.c_double, ctypes.c_double, ctypes.c_double,
    ctypes.POINTER(ctypes.c_double), ctypes.POINTER(ctypes.
        c_double),
    ctypes.POINTER(ctypes.c_double), ctypes.POINTER(ctypes.
        c_double)
]
```

```
# Inputs
r = 4.0
P = np.array([[7.0], [0.0]])

# Prepare output containers
Ax = ctypes.c_double()
Ay = ctypes.c_double()
Bx = ctypes.c_double()
By = ctypes.c_double()

# Call the C function
lib.tangent_points(r, float(P[0]), float(P[1]),
                  ctypes.byref(Ax), ctypes.byref(Ay),
                  ctypes.byref(Bx), ctypes.byref(By))

# Convert results to numpy arrays
O = np.array([[0.0], [0.0]])
A = np.array([[Ax.value], [Ay.value]])
```

```
B = np.array([[Bx.value], [By.value]])

print(Tangent Points from C library:)
print(fA({A[0,0]:.2f}, {A[1,0]:.2f}))
print(fB({B[0,0]:.2f}, {B[1,0]:.2f}))
# Plot using your libs
x_circ = circ_gen(0, r)
x_tan1 = line_gen(P, A)
x_tan2 = line_gen(P, B)
x_OP = line_gen(0, P)
plt.plot(x_circ[0, :], x_circ[1, :], label='Circle')
plt.plot(x_tan1[0, :], x_tan1[1, :], color='orange', label='
    Tangent 1')
plt.plot(x_tan2[0, :], x_tan2[1, :], color='green', label='
    Tangent 2')
plt.plot(x_OP[0, :], x_OP[1, :], 'k--', label='$OP$')
# Mark and label points
points = {'O': 0, 'P': P, 'A': A, 'B': B}
for name, pt in points.items():
```

```
x, y = float(pt[0]), float(pt[1])
plt.scatter(x, y, s=45, zorder=5)
plt.text(x + 0.25, y + 0.25, f'{name}({round(x,2)}, {round(y,2)})',
         fontsize=9, ha='center')
# -style axes
ax = plt.gca()
ax.spines['top'].set_color('none')
ax.spines['right'].set_color('none')
ax.spines['left'].set_position('zero')
ax.spines['bottom'].set_position('zero')
plt.axis('equal')
plt.grid(True)
plt.xlabel('$x$')
plt.ylabel('$y$')
plt.title('Tangents from External Point (using C library)')
plt.legend(loc='upper right')
plt.show()
```

Plot

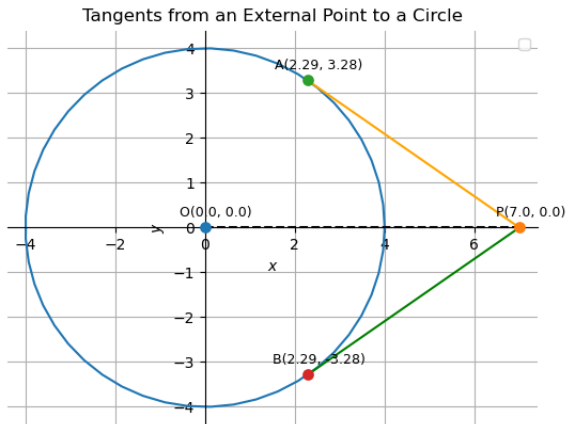


Figure: